

Module – Introduction to firewallld

1. Install firewallld on your Linux system, start the service, and verify its status using systemctl and firewall-cmd commands.

ANS :

- ❖ Install firewallld (RHEL/CentOS/Rocky Linux):
 - sudo yum install firewallld -y
 - ❖ Start and Enable firewallld:
 - sudo systemctl start firewallld
 - sudo systemctl enable firewallld
 - ❖ Check Service Status:
 - sudo systemctl status firewallld
 - ❖ Verify Using firewall-cmd:
 - sudo firewall-cmd --state
2. Add a new rule to firewallld to allow incoming HTTP (port 80) traffic, then use firewall-cmd to confirm the rule is active.
Hint: Use the --permanent and --reload options to make changes persistent.

ANS :

- ❖ Add HTTP Service Permanently:
 - sudo firewall-cmd --permanent --add-service=http
 - ❖ Reload Firewall Rules:
 - sudo firewall-cmd --reload
 - ❖ Verify Rule:
 - sudo firewall-cmd --list-services
3. Temporarily block all incoming traffic on port 8080 using firewallld, then test by attempting to access a simple web server (e.g., Python's http.server) running on that port.

ANS :

- ❖ Start a Test Web Server:
 - python3 -m http.server 8080
- ❖ Block Port 8080:
 - sudo firewall-cmd --permanent --remove-port=8080/tcp
 - sudo firewall-cmd --reload
- ❖ Reject Traffic Explicitly:

- `sudo firewall-cmd --permanent --add-rich-rule='rule family="ipv4" port port="8080" protocol="tcp" reject'`
`sudo firewall-cmd --reload`
 - ❖ Test Access:
 - `curl http://localhost:8080`
 - ❖ From another machine:
 - `http://<server-ip>:8080`
4. List all currently active firewalld zones and display the services and ports allowed in the public zone.

ANS :

- ❖ Show Active Zones:
 - `sudo firewall-cmd --get-active-zones`
 - ❖ Display Public Zone Configuration:
 - `sudo firewall-cmd --zone=public --list-all`
 - ❖ ANS:
 - Active Zone: public
 - Allowed Services:
ssh http
 - Allowed Ports:
(Displays configured ports, if any)
5. Remove a custom rule you added earlier (such as the HTTP port 80 rule) and verify that the change has taken effect using firewall-cmd.

ANS :

- ❖ Remove HTTP Service:
 - `sudo firewall-cmd --permanent --remove-service=http`
- ❖ Reload Configuration:
 - `sudo firewall-cmd --reload`
- ❖ Verify:
 - `sudo firewall-cmd --list-services`
- The HTTP rule was removed successfully.
The 'http' service no longer appears in the allowed services list.